Can a small local agent redo a cloud agent's work?

93 evaluator-verified tasks from an agent-built health-data repository, with an open harness and a Gemma 4 E4B baseline

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Gemma 4 Developer Agent Competition, Paper Track [17].
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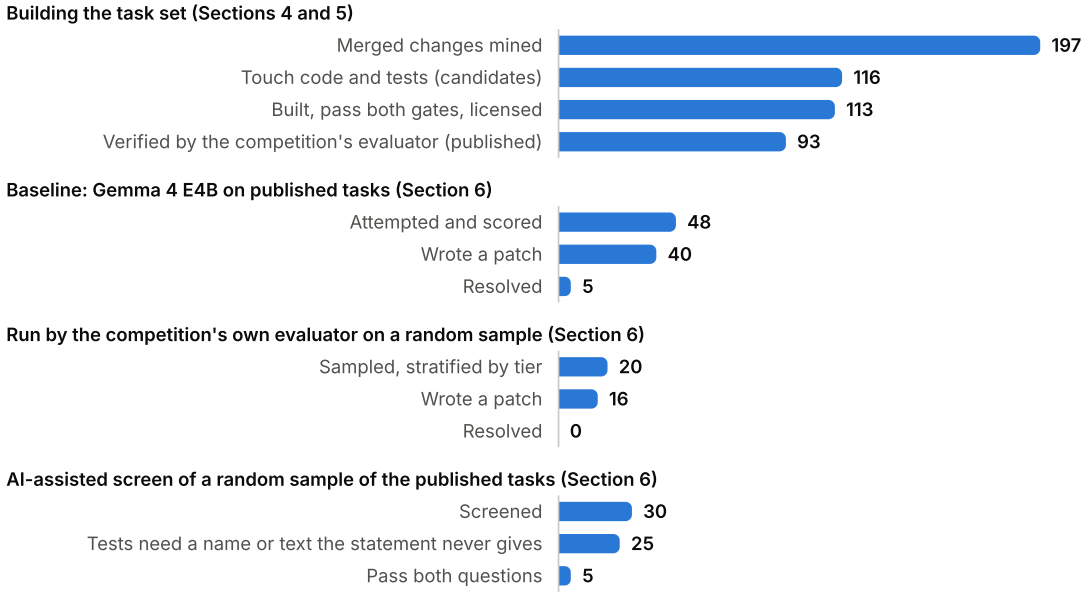
A personal learning project, not affiliated with or endorsed by any employer, institution or vendor. Every patient record in the source repository is synthetic.

ABSTRACT

Coding-agent benchmarks are mined from mature web libraries maintained by people. I release a task set mined from a different kind of repository: a health-data benchmark built by cloud coding-agent sessions under written rules that CI enforces (960 commits in fifteen days, 775 of them marked as agent-authored). It has 93 tasks in the competition's format, each verified with the competition's own evaluator and labelled with where its problem statement came from: an issue, a pull request or a commit message. Under tight limits (2,048 output tokens, thinking off), Gemma 4 E4B on two free T4 GPUs resolved 5 of 48 tasks, run cheapest first. A paired rerun of 23 with the code-graph tools resolved 2, as without them; the model rarely called the tools. An AI-assisted screen of 30 finds 25 with tests needing names the statement never gives. The agent ran on my open re-implementation of the competition's harness; the official evaluator gives the same verdict on all 48 patches compared. Run end to end by that evaluator on a seeded sample of 20, with the competition's hardware and my competition settings, the agent resolved none. I report how the model fails and what each task costs.

Data and code. Tasks (CC BY 4.0): kaggle.com/datasets/jcblearning/health-transformation-lab-coding-agent-tasks.
Harness, builders and run tables (Apache 2.0): github.com/healthtransformationlab/health-transformation-lab-nlp.
Agent configuration and each number's source (Apache 2.0): github.com/healthtransformationlab/gemma-developer-agent.

Figure 1. Task counts at each step, on one scale.



1. Introduction

The best coding agents today run on large models in the cloud. This competition [17] asks whether a model that fits on one accelerator can do the same work. The tasks available come from a narrow place: SWE-bench draws on twelve popular Python libraries [1], and the competition's development set on four.

Two kinds of repository are scarce. In a **data repository** the costly defect is silent: a pipeline finishes, a table is written, and a number in it is wrong (Section 7). An **agent-built repository**, written by agents under a person's direction, can leave an unusually complete record: an issue, a pull request and a review for each change. A large cloud agent did this work. How much of it can a small local agent redo from the same problem statements?

I study one repository that is both. `health-transformation-lab-nlp` is part of Health Transformation Lab, my open health-data benchmarks. Its job is to surface silent defects, here in how well gum disease can be found in synthetic dental records. I typed no code: every change was made by coding-agent sessions I directed in chat and merged after an automated review.

Contributions:

1. **A task set** of 93 tasks in the competition's format, each verified with the competition's own evaluator, public on Kaggle under CC BY 4.0, with a fidelity tier per task (Section 4).
2. **An open harness checked against the official one**: a sandboxed re-implementation that agrees with the competition's evaluator on every saved agent patch and keeps answers from the model (Section 5).
3. **A baseline** for Gemma 4 E4B with a failure breakdown, cost per task, a paired test of the code-graph tools, and a run under the competition's own evaluator (Section 6).
4. **A check on a second repository**: without the tracing rule, the same mining rule finds candidates but no issue-linked statements (Section 4.5).

2. Related work

Task sets from repository history. SWE-bench pairs GitHub issues with the pull requests that closed them and scores a patch by the repository's tests [1]; SWE-bench Verified adds human screening of task quality [2]. Defects4J [3] and BugsInPy [4] did the same for Java and Python bugs before language models. SWE-Gym [5], SWE-smith [6] and SWE-rebench [7] scale task collection for training and for contamination-free evaluation. All draw on human-maintained libraries; this set applies the recipe to an agent-built data repository and labels each statement's source.

Agents. SWE-agent showed that the interface matters as much as the model [8]; the competition fixes it.

Data work. DS-1000 tests data-science code generation [9]; Spider 2.0 [10], ELT-Bench [11] and data-eng-bench [16] score an artefact agents produce: SQL, a pipeline, dbt models. MLOps-Bench [15] is closest: 212 fail-to-pass tasks from 31 public ML repositories.

Health-data quality. The repository's checks follow the harmonized data-quality terms of Kahn et al. [12]; its de-identification tests follow the i2b2/UTHealth categories [13] and use MACCROBAT [14].

3. The source repository

The repository holds a Python package and tests, an R twin kept in numerical parity, SQL and Databricks pipelines, and documentation, written October 25 to November 8, 2026.

Who wrote it. Of 960 commits on `main`, 775 are marked as agent-authored, 175 are merge commits GitHub made for pull requests, 5 come from an automation bot, 4 carry only my name, and one is the initial commit.

Rules the agents work under. A checked-in `AGENTS.md` requires an issue per user story, a branch and pull request tied to it, and a test before the code. CI rejects a pull request from an agent's own session branch or without an issue link.

Reviews. Every pull request has a review committed as a file: 240 of them. Of the 192 with a structured verdict, all approve (64 with minor findings, 9 with conditions), since blocking findings are fixed first. Agents merge their own green pull requests; runs on limited compute, publishing, deletions and rule changes wait for me.

4. Building the task set

4.1 Mining

The mining script reads every merged pull request and every first-parent commit on `main` as of October 29, 2026: 197 changes. A change is a **candidate** when its diff touches both non-test code and test files, which is the competition's own mining rule.

Step	Tasks
Merged changes	197
Touch code and tests (candidates)	116
Built (one withdrawn: its patch names an organisation the repository's rules forbid)	115
Pass both gates	114
One withheld (its patch adds third-party files with no stated licence)	113
Published: the competition's evaluator resolves the reference patch (Section 5)	93

4.2 Fidelity tiers

Each task records where its `problem_statement` came from.

Tier	Statement source	Candidates (published)
A	the issue body	49 (39)
B	the pull request's title and body (no issue existed)	45 (34)
C	the commit message (no pull request existed)	22 (20)

Tiers B and C exist because the issue rule was adopted partway through the history. For those 67 candidates a **backfilled issue record** quotes the original text verbatim, carries a `backfilled` label and is closed on creation.

4.3 Snapshots, graphs, embeddings

A snapshot is the tree at `base_commit` as a single commit, so no later history comes along. The code graph is an AST walk over the package into NetworkX node-link JSON with the competition's keys (median 488 nodes and 397 `calls` edges). Embeddings come from EmbeddingGemma, cut to 256 dimensions and renormalised. A format check against the competition's own files matched the snapshot layout, graph keys and embedding shape; the competition does not name its embedding model.

4.4 Gates

A task is kept only if its tests fail at `base_commit` with the test patch alone, and pass once the reference patch is applied with the files the grader strips removed. All 115 fail without the fix; 114 pass with it.

4.5 A second repository

To see whether the recipe transfers, I applied the mining rule unchanged to a second public repository: Open Navigator, run by a nonprofit I lead, with several human contributors and no issue-tracing rule. Of 1,320 first-parent commits, 133 touch non-test Python and Python tests. None links an issue. Only 22 can be tied by commit subject to one of the repository's 169 merged pull requests.

So the rule finds candidates anywhere, but statements depend on what the history recorded: 67 of this paper's 116 candidates got their issue record afterwards. These 133 are not gated and not released.

5. Harness and verification

My harness. The competition's guide documents the initial prompt, the multi-turn loop, the nine tools with their limits, the budgets, patch extraction and verification. I re-implemented those sections before I found the competition's evaluator, which ships in a separate Kaggle dataset. The prompt, nudge messages and budget warning follow the guide's wording. Two scripted models bound the harness. Replaying each reference patch from the repository's history resolves 111 of 113 tasks (the other two need Postgres and R), and submitting nothing resolves none.

The competition's evaluator. I then ran the competition's own verification (`swegemma` 0.2.11, subprocess sandbox, unmodified) on the 113 tasks first published and on the corrected release.

Check	Result
First release as published, empty patch (3 tasks)	all 3 fail at import, before any test runs
One file added per snapshot, reference patch	93 of 113 resolved
One file added per snapshot, empty patch	0 of 113 resolved
One file added per snapshot, Gemma's saved patches	same verdict as my harness on 25 of 25
Corrected release as published, reference patch	93 of 93 resolved
Corrected release as published, empty patch	0 of 93 resolved
Corrected release as published, Gemma's rerun patches	same verdict as my harness on 23 of 23

The first release could not be scored: the evaluator puts the repository root on the import path, and this package sits in `python/`. The corrected release adds a short `conftest.py` at each snapshot's root and keeps only the 93.

The 20 left out fall into three groups. Twelve have tests that read third-party files the published snapshots withhold for licence reasons, so they cannot pass from a snapshot under any verifier; my control missed this because it ran from the repository's history. Six start a script that imports a module beside it, which the evaluator's `PYTHONSAFEPATH=1` forbids and my harness allows. Two need Postgres or R. Elsewhere the two verifiers agree on every reference patch.

Keeping answers from the model. This repository's history contains every fix, and later snapshots contain earlier tasks' fixes. Commands and verification run in Linux user, mount, network and pid namespaces with no network and an allow-listed environment. A review before the first pull request found that an earlier version ran the model's commands on the host, where it could read every reference patch.

Kaggle GPUs do not allow namespaces, and a notebook can read every attached file, so each task runs alone in a private, offline notebook beside a private dataset holding that task only; the patch is verified locally.

Known differences. Namespaces instead of a container; no ADK context compaction or prefix caching; sub-agents and skills are not run; `PYTHONSAFEPATH` is not set. No number here is a competition score.

6. Baseline

Setup. `gemma-4-e4b-it` in fp16 with transformers on two T4 GPUs, internet off, thinking off, at most 2,048 new tokens per turn, a 24,576-token context, 40 tool calls, 60 turns and 30 minutes per task. Tasks were run cheapest first under a GPU-hour cap.

Results. Ten of the 58 tasks attempted are among the twenty left out (Section 5), so they are set aside. On the other 48 the model wrote a patch for 40 and resolved 5 (10.4%, 95% Wilson CI 4.5–22.2). By tier, it resolved 4 of 24 tasks whose statement is an issue, 1 of 19 from a pull request and 0 of 5 from a commit message. The intervals overlap.

Do the graph tools help? A harness fault served the first tasks without their code graph, so the graph tools had nothing to query. After fixing it I reran 28 of them with the same settings; the 23 published give a paired comparison.

23 tasks, each run twice	Non-empty patch	Resolved (95% CI)
Graph tools unavailable	18	2 (8.7%, 2.4–26.8)
Graph tools available	20	2 (8.7%, 2.4–26.8)

One task was resolved only without the tools, one only with them, one both times. The transcripts suggest why: with the tools available the model called one in 4 of the 23 runs, 6 calls among 145. It reads and edits files and rarely searches. The figures use each task's latest run; eight tasks have none with the tools.

How it fails. Of the 43 unresolved tasks, 35 submitted a patch that fails the target tests (30 with failing tests, 5 with errors before any test ran) and 8 submitted nothing. In those eight transcripts, four have a turn that hit the 2,048-token limit, three submit without an edit, and one calls the edit tool four times with an argument it does not take. The five resolved tasks are small: a CI rule, a warning message, a label parser, a training setting, a notebook setting.

Task fairness. An AI reviewer screened 30 of the 93 on SWE-bench Verified's two questions [2]; I checked its findings in summary and agreed with all 30. Five pass. In 25 the tests need a name or text the statement never gives. Of 16 the model attempted, it resolved 1 of the 3 that pass and none of the 13 that fail, so the baseline may measure the tasks more than the model.

Under the competition's evaluator. I also ran the agent end to end with the official harness: `swegemma`'s agent loop, vLLM from the competition's wheelhouse, four L4 GPUs, and my competition settings (16,384 output tokens, a 4,096-token thinking budget, 40 tool calls, 5 minutes). The sample is 20 of the 93, drawn with a fixed seed and stratified by tier, not by cost. Each task ran alone, without its patch or tests; the same evaluator then verified the patch on the published snapshot. It resolved 0 of 20 (95% CI 0–16.1): 16 patches fail and 4 runs submit nothing. No run reached a budget. In 189 tool calls the model called a graph tool 9 times, the analyzer sub-agent 3

times and the skill never. Thirteen of the 20 are also in the screen above; 11 of them fail it. The run also found a defect my harness had hidden: my prompt required a hints variable, which ends the agent on the half of the tasks that have no hints. I fixed it before those tasks ran.

Cost. The 58 first attempts used 20.6 GPU-hours and the 28 reruns 9.6: 30.2 in all, about $21 at an estimated $0.69 per T4-hour; Kaggle's GPUs are free. The median task takes 0.39 GPU-hours. Five resolved tasks cost 6.0 GPU-hours each. The evaluator's 21 runs used 12.8 L4 GPU-hours.

7. What these tasks contain that web-library tasks do not

Three cases show the defect class that motivates the set: everything ran, and a number was wrong.

Case	Score on what it was built against	Score on what it was not
De-identification rules, share of identifiers removed	1.00 (the tier the rules were written beside)	0.27 (a tier written after the rules were frozen)
Tooth-loss model, AUC	0.89–0.90 (held-out labelled extractions)	0.59–0.63 (teeth missing at the first visit)
A masking defect in the feature table	every test passed	the measures of about a quarter of charted teeth were blank in every earlier run

8. Limitations

- **One repository, one author's rules.** The second repository tests only the mining rule.
- **Agent-written statements.** The issue text was largely written by a cloud agent at my direction, which may make it easier or harder to follow than a person's.
- **My harness ran the baseline.** The competition's evaluator ran the agent on 20 of my tasks only, so no score here is comparable to the leaderboard.
- **Constrained settings.** The baseline caps output at 2,048 tokens with thinking off on T4 GPUs. Only the 20 tasks under the evaluator ran with my competition settings, which allow 5 minutes and 40 tool calls against the default 60 and 100.
- **Small.** 48 of 93 tasks, cheapest first, and a random 20 under the evaluator; one sampled run per condition at temperature 0.2. One task flipping moves the paired result.
- **A screen checked only in summary.** I saw the AI's verdict first and did not read the tests; 63 tasks are unscreened.
- **The tasks are not silent.** Each comes with a failing test. The set measures repair of the code that looks for silent defects, not noticing one unprompted.
- **Synthetic health data.** The cases in Section 7 are agreement with a synthetic answer key, not findings about any patient or practice.

9. Conclusion and release

A cloud agent built this repository in fifteen days. From the same problem statements, a model that runs on free GPUs resolved 5 of 48 tasks, all of them small, and none of a random 20 under the competition's own evaluator; it left the graph tools almost unused. The released tasks let others test other models and settings.

The recipe should transfer to any repository that traces changes to issues: mine changes that touch code and tests, record each statement's source, gate, run reference-patch and empty-patch controls on the published files with the official evaluator, and screen statements against tests before trusting any model's score.

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